

Great Scott Gadgets

HackRF Pro

HackRF Pro from Great Scott Gadgets is a Software Defined Radio peripheral capable of transmission or reception of radio signals from 100 kHz to 6 GHz. Designed to enable test and development of modern and next generation radio technologies, HackRF Pro is an open source hardware platform that can be used as a USB peripheral or programmed for stand-alone operation.



- 100 kHz to 6 GHz operating frequency
- Tunable from 0 Hz to 7.1 GHz
- Half-duplex transceiver
- Up to 20 million samples per second
- 8-bit quadrature samples (8-bit I and 8-bit Q)
- Compatible with GNU Radio, SDR#, and more
- Software-configurable RX and TX gain and baseband filter
- Software-controlled RF port power (50 mA at 3.3 V)
- SMA RF connector
- SMA clock input and output for synchronization and triggering
- Convenient buttons for programming
- Internal pin headers for expansion
- High-Speed USB 2.0 with Type-C connector
- USB-powered
- Open source hardware

Compared to [HackRF One](#), HackRF Pro introduces a host of new and updated features, including:

- Wider operating frequency range
- Improved RF performance with flatter frequency response
- Modern USB Type-C connector
- Built-in TCXO crystal oscillator for superior timing stability
- Logic upgrade from a CPLD to a power-efficient FPGA
- Elimination of the DC spike
- Extended-precision mode with 16-bit samples for low sample rates (typical ENOB: 9-11)
- Half-precision mode with 4-bit samples at up to 40 Msps
- More RAM and flash memory for custom firmware
- Installed shielding around the radio section
- Trigger input and output accessible through clock connectors
- Cutout in the PCB provides space for future add-ons
- Improved power management
- Enhanced RF port protection
- Facility to hardware-disable transmit mode

Software that works with HackRF One is already compatible with HackRF Pro. We designed HackRF Pro for backward compatibility, following the same basic architecture of HackRF One but with many small enhancements. Prior to shipping HackRF Pro, we will publish a migration guide that will show software developers how to take advantage of certain new capabilities of HackRF Pro, but out-of-the-box HackRF Pro will behave like HackRF One with superior performance. In addition to host software compatibility, our migration guide will address firmware, allowing developers to port custom firmware to the new platform and take advantage of its unique capabilities

HackRF Pro has an injection molded plastic enclosure. USB cable and antenna are not included. [ANT500](#) is recommended as a starter antenna for HackRF Pro.